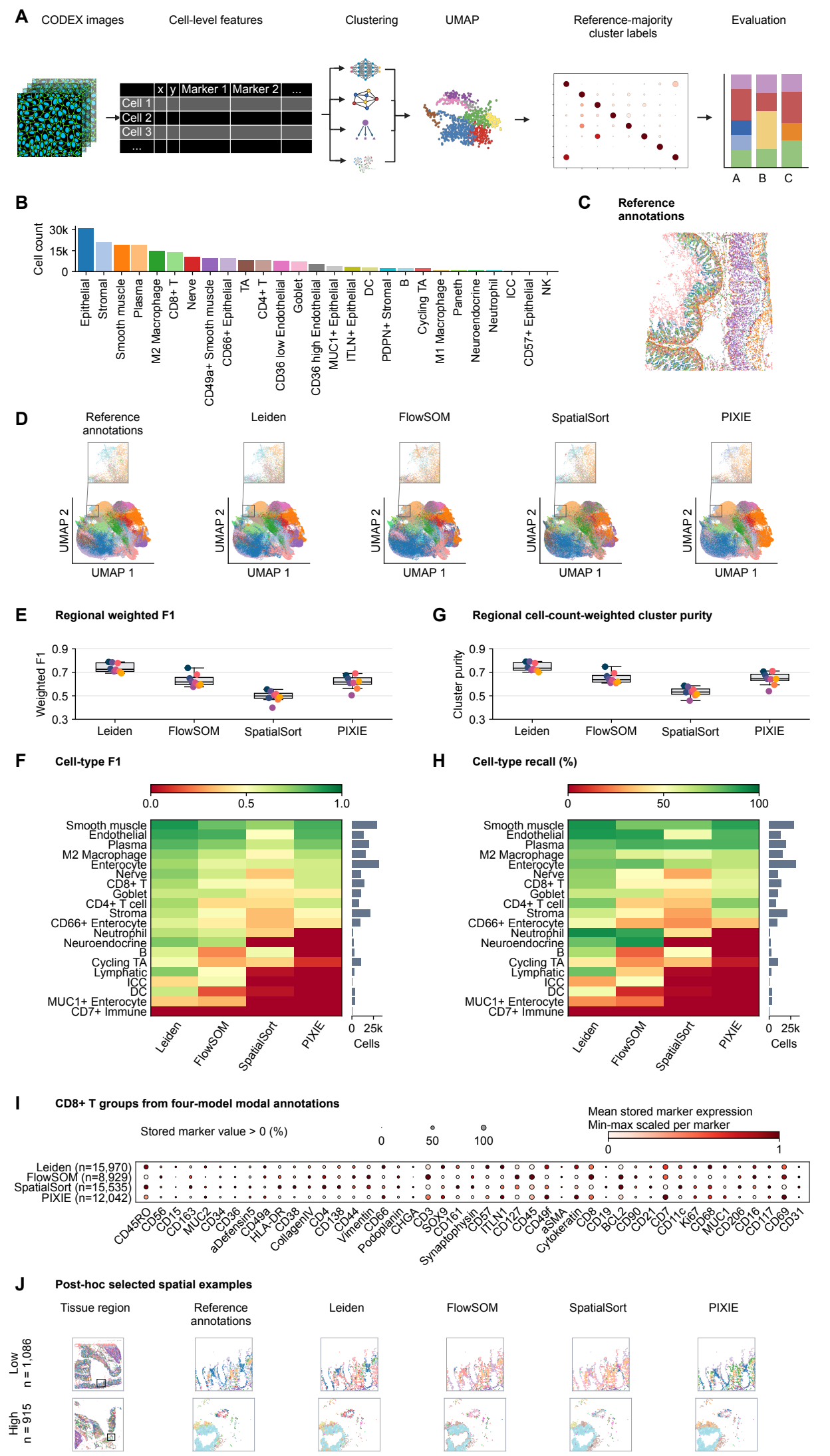

A
CODEX images
Cell-level features
Clustering
UMAP
Reference-majority cluster labels
Evaluation
x
y
Marker 1
Marker 2
...
Cell 1
Cell 2
Cell 3
A
B
C
B
Cell count
30k
15k
0
Epithelial
Stromal
Smooth muscle
Plasma
M2 Macrophage
CD8+ T
Nerve
CD49a+ Smooth muscle
CD66+ Epithelial
TA
CD4+ T
CD36 low Endothelial
Goblet
CD36 high Endothelial
MUC1+ Epithelial
ITLN+ Epithelial
DC
PDPN+ Stromal
B
Cycling TA
M1 Macrophage
Paneth
Neuroendocrine
Neutrophil
ICC
CD57+ Epithelial
NK
C
Reference annotations
D
Reference annotations
Leiden
FlowSOM
SpatialSort
PIXIE
UMAP 2
UMAP 1
E
Regional weighted F1
Weighted F1
0.9
0.7
0.5
0.3
Leiden
FlowSOM
SpatialSort
PIXIE
G
Regional cell-count-weighted cluster purity
Cluster purity
0.9
0.7
0.5
0.3
Leiden
FlowSOM
SpatialSort
PIXIE
F
Cell-type F1
0.0
0.5
1.0
Smooth muscle
Endothelial
Plasma
M2 Macrophage
Enterocyte
Nerve
CD8+ T
Goblet
CD4+ T cell
Stroma
CD66+ Enterocyte
Neutrophil
Neuroendocrine
B
Cycling TA
Lymphatic
ICC
DC
MUC1+ Enterocyte
CD7+ Immune
Leiden
FlowSOM
SpatialSort
PIXIE
0
25k
Cells
H
Cell-type recall (%)
0
50
100
Smooth muscle
Endothelial
Plasma
M2 Macrophage
Enterocyte
Nerve
CD8+ T
Goblet
CD4+ T cell
Stroma
CD66+ Enterocyte
Neutrophil
Neuroendocrine
B
Cycling TA
Lymphatic
ICC
DC
MUC1+ Enterocyte
CD7+ Immune
Leiden
FlowSOM
SpatialSort
PIXIE
0
25k
Cells
I
CD8+ T groups from four-model modal annotations
Stored marker value > 0 (%)
0
50
100
Mean stored marker expression
Min-max scaled per marker
0
1
Leiden (n=15,970)
FlowSOM (n=8,929)
SpatialSort (n=15,535)
PIXIE (n=12,042)
CD45RO
CD56
CD15
CD163
MUC2
CD34
CD36
aDefensin5
CD49a
HLA-DR
CD38
CollagenIV
CD4
CD138
CD44
Vimentin
CD66
Podoplanin
CHGA
CD3
SOX9
CD161
Synaptophysin
CD57
ITLN1
CD127
CD45
CD49f
aSMA
Cytokeratin
CD8
CD19
BCL2
CD90
CD21
CD7
CD11c
Ki67
CD68
MUC1
CD206
CD16
CD17
CD69
CD31
J
Post-hoc selected spatial examples
Tissue region
Reference annotations
Leiden
FlowSOM
SpatialSort
PIXIE
Low
n = 1,086
High
n = 915